Safety Data Sheet
Basolon GL 40 exp

1. Identification

Product identifier used on the label

Basolon GL 40 exp

Recommended use of the chemical and restriction on use

Recommended use*: Performance Chemicals for Oilfield Applications
Recommended use*: for industrial use only
Use as an intermediate; Use as Monomer

* The “Recommended use” identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:
BASF Canada Inc.
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Mississauga, ON, L4W 0B6, CANADA
Global Oilfield Chemicals

Telephone: +49 621 60-0
E-mail address: RegXcellenceOilfieldChemicals@basf.com

Emergency telephone number
24 Hour Emergency Response Information
CHEMTREC: 1-800-424-9300
BASF HOTLINE: (800) 454-COPE (2673)

Other means of identification

Chemical family:	aldehydes, in water
Synonyms:	Glyoxal Use: Chemical used in synthesis and/or formulation of industrial products.

2. Hazards Identification

According to Hazardous Products Regulations (HPR) (SOR/2015-17)

Classification of the product

Acute Tox.	4 (Inhalation - mist)	Acute toxicity
Skin Irrit.	2	Skin irritation
Eye Irrit.	2B	Eye irritation
Skin Sens.	1	Skin sensitization
STOT SE	3 (irritating to respiratory system)	Specific target organ toxicity — single exposure
Muta.	2	Germ cell mutagenicity
STOT RE	2	Specific target organ toxicity — repeated exposure
Aquatic Acute	3	Hazardous to the aquatic environment - acute

Label elements

Pictogram:

Signal Word:
Warning

Hazard Statement:

H320	Causes eye irritation.
H315	Causes skin irritation.
H332	Harmful if inhaled.
H317	May cause an allergic skin reaction.
H335	May cause respiratory irritation.
H341	Suspected of causing genetic defects.
H373	May cause damage to organs (Kidney) through prolonged or repeated exposure.
H402	Harmful to aquatic life.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P271	Use only outdoors or in a well-ventilated area.
P260	Do not breathe dust/gas/mist/vapours.
P201	Obtain special instructions before use.
P273	Avoid release to the environment.
P202	Do not handle until all safety precautions have been read and understood.
P272	Contaminated work clothing should not be allowed out of the workplace.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P308 + P313	IF exposed or concerned: Get medical attention.
P302 + P352	IF ON SKIN: Wash with plenty of soap and water.
P333 + P313	If skin irritation or rash occurs: Get medical attention.
P332 + P313	If skin irritation occurs: Get medical attention.
P362 + P364	Take off contaminated clothing and wash it before reuse.
P337 + P313	If eye irritation persists: Get medical attention.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents/container in accordance with local regulations.

Hazards not otherwise classified

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

Labeling of special preparations (GHS):
This product is capable of releasing formaldehyde into the air. May cause cancer.

3. Composition / Information on Ingredients

According to Hazardous Products Regulations (HPR) (SOR/2015-17)

glyoxal
CAS Number: 107-22-2
Content (W/W): >= 39.5 - <= 40.5%
Synonym: Glyoxal

ethylene glycol
CAS Number: 107-21-1
Content (W/W): >= 0.0 - <= 2.5%
Synonym: 1,2-Ethanediol; Ethylene glycol

Acetic acid, hydroxy-
CAS Number: 79-14-1
Content (W/W): >= 0.05 - <= 0.35%
Synonym: Hydroxyacetic acid; Glycolic acid

4. First-Aid Measures

Description of first aid measures

General advice:
Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air. Assist in breathing if necessary. Seek medical attention.

If on skin:
Wash affected areas thoroughly with soap and water. Remove contaminated clothing. If irritation develops, seek medical attention.

If in eyes:
In case of contact with the eyes, rinse immediately for at least 15 minutes with plenty of water. Immediate medical attention required.

If swallowed:
Rinse mouth and then drink 200-300 ml of water. Do not induce vomiting. Immediate medical attention required.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Information on: glyoxal
Symptoms: Overexposure may cause:, Eye irritation, skin irritation, erythema, nausea, headache, vomiting, dizziness, diarrhea, abdominal cramps
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Indication of any immediate medical attention and special treatment needed

Note to physician

Treatment:	Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, foam, dry powder, carbon dioxide

Unsuitable extinguishing media for safety reasons:
water jet

Special hazards arising from the substance or mixture
Hazards during fire-fighting:
nitrogen oxides, carbon oxides
The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters
Protective equipment for fire-fighting:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

Impact Sensitivity:
Remarks: Based on the chemical structure there is no shock-sensitivity.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures
Avoid contact with the skin, eyes and clothing.

Environmental precautions
Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up
For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr). Dispose of absorbed material in accordance with regulations.

7. Handling and Storage

Precautions for safe handling
Ensure thorough ventilation of stores and work areas.

Conditions for safe storage, including any incompatibilities

Suitable materials for containers: Stainless steel 1.4401, Stainless steel 1.4301 (V2), High density polyethylene (HDPE), glass, Low density polyethylene (LDPE)
Unsuitable materials for containers: Paper/Fibreboard

Further information on storage conditions: Protect from air.

Storage stability:
Storage temperature: < 50 °C
Storage duration: 6 Months
May yellow after lengthy storage.
During the storage a product characteristic clouding or crystallin precipitation of the trimeric glyoxal hydrate may occur.
The process is reversible if warmed up mild (max. 40 °C).
From the data on storage duration in this safety data sheet no agreed statement regarding the warrantee of application properties can be deduced.
Protect from temperatures above: 50 °C
The packed product must be protected against exceeding the indicated temperature.

8. Exposure Controls/Personal Protection

<u>Components with occupational exposure limits</u>

ethylene glycol	ACGIH, US:	TWA value 25 ppm Vapor fraction ;
	ACGIH, US:	STEL value 50 ppm Vapor fraction ;
	ACGIH, US:	STEL value 10 mg/m3 Inhalable fraction, Aerosole ;

glyoxal	ACGIH, US:	TWA value 0.1 mg/m3 Inhalable fraction and vapor ;

Advice on system design:
Provide local exhaust ventilation to control vapours/mists.

Personal protective equipment

Respiratory protection:
Do not exceed the maximum use concentration for the respirator facepiece/cartridge combination. For emergency or non-routine, high exposure situations, use a NIOSH-certified full facepiece pressure demand self-contained breathing apparatus (SCBA) or a full facepiece pressure demand supplied-air respirator (SAR) with escape provisions. Wear a NIOSH-certified (or equivalent) organic vapour respirator.

Hand protection:
Chemical resistant protective gloves, Suitable materials may include, fluoroelastomer (Viton), butyl rubber, chloroprene rubber (Neoprene), nitrile rubber (Buna N), polyvinylchloride (Pylox), Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Tightly fitting safety goggles (chemical goggles). Wear face shield if splashing hazard exists.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. head protection, apron, protective boots, chemical-protection suit.

General safety and hygiene measures:
Avoid contact with the skin, eyes and clothing. When using, do not eat, drink or smoke. Avoid contact with skin and eyes. Remove soiled or soaked clothing immediately. Wash off any contamination that gets onto the skin with plenty of water and soap, skin care.

9. Physical and Chemical Properties

Form:	liquid	
Odour:	aldehyde-like	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
Colour:	colourless to yellow	
pH value:	2 - 3 (40 %(m), 20 °C)	
melting range:	-50 - -15 °C	(OECD Guideline 102)
Freezing point:	No data available.	
Boiling point:	103.6 °C (1,013.25 hPa)	(Directive 92/69/EEC, A.2)
Flash point:	> 93 °C	(ISO 13736)
Flammability:	not flammable	(derived from flash point)
Lower explosion limit:	(103 °C, 1013 hPa) Not determinable. For liquids not relevant for classification and labelling.	(air)

Upper explosion limit:	For liquids not relevant for classification and labelling. Not determinable.	
Autoignition:	285 °C	(DIN 51794)
Vapour pressure:	20.2 hPa (20 °C) Information applies to the solvent.	(Directive 92/69/EEC, A.4)
	106.7 hPa (50 °C) Information applies to the solvent.	(Directive 92/69/EEC, A.4)
Density:	approx. 1.27 g/cm3 (40 %(m), 20 °C)	(Directive 92/69/EEC, A.3)
	1.2514 g/cm3 (50 °C)	
Relative density:	1.27 (20 °C)	(Directive 92/69/EEC, A.3)
Vapour density:	Water content greater than 10%.	
Partitioning coefficient n-octanol/water (log Pow):	-1.15 (23 °C)	(Directive 92/69/EEC, A.8)
	-1 (23 °C)	(Directive 84/449/EEC, A.8)
	-1.62 (23 °C)	(Directive 84/449/EEC, A.8)
Self-ignition temperature:	Based on its structural properties the product is not classified as self-igniting.	
Thermal decomposition:	approx. > 120 °C No decomposition if correctly stored and handled.	
Viscosity, dynamic:	8.37 mPa.s (20 °C)	(OECD Guideline 114)
	4.25 mPa.s (40 °C)	(OECD Guideline 114)
Viscosity, kinematic:	6.6 mm2/s (20 °C)	(OECD Guideline 114)
	3.38 mm2/s (40 °C)	(OECD Guideline 114)
Particle size:	The substance / product is marketed or used in a non solid or granular form.	
Solubility in water:	(20.2 °C) miscible	
Molar mass:	58.04 g/mol	
Evaporation rate:	Value can be approximated from Henry's Law Constant or vapor pressure.	

10. Stability and Reactivity

Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals:
No corrosive effect on metal.

Oxidizing properties:
Based on its structural properties the product is not classified as oxidizing.

Formation of flammable gases:	Remarks:	Forms no flammable gases in the presence of water.

Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions
Reacts with strong alkalies. Exothermic reaction.

Conditions to avoid
Temperature: > 50 degrees Celsius
See SDS section 7 - Handling and storage.

Incompatible materials
strong alkalies

Hazardous decomposition products

Decomposition products:
Hazardous decomposition products: No hazardous decomposition products if stored and handled as prescribed/indicated.

Thermal decomposition:
approx. > 120 °C

No decomposition if correctly stored and handled.

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity
Assessment of acute toxicity: Of low toxicity after single ingestion. Virtually nontoxic after a single skin contact. Of moderate toxicity after short-term inhalation.

Oral
Type of value: LD50
Species: rat (male/female)
Value: 3,300 mg/kg (OECD Guideline 401)

Inhalation
Type of value: LC50
Species: rat (male/female)
Value: 2.44 mg/l (OECD Guideline 403)
Exposure time: 4 h
An aerosol was tested.

Dermal

Type of value: LD50
Species: rat (male/female)
Value: > 2,000 mg/kg (OECD Guideline 402)
Limit concentration test only (LIMIT test).

Assessment other acute effects
Assessment of STOT single:
Causes temporary irritation of the respiratory tract.

Irritation / corrosion
Assessment of irritating effects: Eye contact causes irritation. Skin contact causes irritation.

Skin
Species: rabbit
Result: Irritant.
Method: OECD Guideline 404

Eye
Species: rabbit
Result: Irritant.
Method: OECD Guideline 405

Sensitization
Assessment of sensitization: Caused skin sensitization in animal studies. Caused sensitization in humans.

Guinea pig maximization test
Species: guinea pig
Result: skin sensitizing
Method: OECD Guideline 406

Human Maximization Test
Species: human
Result: skin sensitizing
Method: Patch test
Literature data.

Aspiration Hazard
No aspiration hazard expected.

Chronic Toxicity/Effects

Repeated dose toxicity

Information on: ethylene glycol
Assessment of repeated dose toxicity: The substance may cause damage to the kidney after repeated ingestion. The substance may cause damage to the kidney after repeated skin contact with high doses.

Genetic toxicity
Assessment of mutagenicity: The substance was mutagenic in various test systems with microorganisms and cell cultures; however, these results could not be confirmed in tests with mammals.

Carcinogenicity
Assessment of carcinogenicity: The whole of the information assessable provides no indication of a carcinogenic effect.

Reproductive toxicity
Assessment of reproduction toxicity: The results of animal studies gave no indication of a fertility impairing effect.

Teratogenicity
Assessment of teratogenicity: No indications of a developmental toxic / teratogenic effect were seen in animal studies.

Medical conditions aggravated by overexposure
Data available do not indicate that there are medical conditions that are generally recognized as being aggravated by exposure to this substance/product. See SDS section 11 - Toxicological information.

12. Ecological Information

Toxicity

Aquatic toxicity
Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms. Based on long-term (chronic) toxicity study data, the product is very likely not harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish
LC50 (96 h) > 186 - < 272 mg/l, Leuciscus idus (DIN 38412 Part 15, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Aquatic invertebrates
EC50 (48 h) 161 mg/l, Daphnia magna (Directive 79/831/EEC, static)
Nominal concentration. The ecological data given are those of the active ingredient.

LC50 (96 h) 76.0 mg/l, Americamysis bahia (other, semistatic)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Aquatic plants
EC50 (72 h) > 40 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

EC10 (72 h) > 10 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

EC50 (72 h) 347.1 mg/l (growth rate), Skeletonema costatum (OECD Guideline 201)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

No observed effect concentration (72 h) 118.4 mg/l (growth rate), Skeletonema costatum (OECD Guideline 201)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Chronic toxicity to fish
No observed effect concentration (34 d) 112 mg/l, Pimephales promelas (OPP 72-4 (EPA-Guideline), Flow through.)
The ecological data given are those of the active ingredient. The statement of the toxic effect relates to the analytically determined concentration.

Chronic toxicity to aquatic invertebrates
No observed effect concentration (21 d) 3.19 mg/l, Daphnia magna (OECD Guideline 211, semistatic)
The statement of the toxic effect relates to the analytically determined concentration. The ecological data given are those of the active ingredient.

Assessment of terrestrial toxicity
Toxic effects have been observed in studies with soil living organisms. Toxic effects have been observed in studies with terrestric plants.

Soil living organisms

Toxicity to soil dwelling organisms:
LC50 (14 d) > 135 mg/kg, Eisenia foetida (OECD Guideline 207, artificial soil)
The ecological data given are those of the active ingredient. The details of the toxic effect relate to the nominal concentration.

EC50 (28 d) > 1,054 mg/kg, soil dwelling microorganisms (OECD Guideline 217, natural soil)
The details of the toxic effect relate to the nominal concentration. The data have been calculated from values for a preparation with a lower substance concentration. The ecological data given are those of the active ingredient.

EC50 (28 d) > 1,054 mg/kg, soil dwelling microorganisms (OECD Guideline 216, natural soil)
The details of the toxic effect relate to the nominal concentration. The data have been calculated from values for a preparation with a lower substance concentration. The ecological data given are those of the active ingredient.

Toxicity to terrestrial plants

No observed effect concentration (21 d) 314 mg/kg, Brassica napus (OECD Guideline 208)
The ecological data given are those of the active ingredient. The data have been deduced from values for a preparation or mixture with a lower substance concentration.

Other terrestrial non-mammals
No data available.

Microorganisms/Effect on activated sludge

Toxicity to microorganisms
DIN 38412 Part 8 static
bacterium/EC10 (16 h): 22.8 mg/l
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

OECD Guideline 209 static

activated sludge/EC50 (0.5 h): > 400 mg/l
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Persistence and degradability

Assessment biodegradation and elimination (H2O)
Readily biodegradable (according to OECD criteria).

Elimination information

90 - 100 % DOC reduction (19 d) (OECD 301 A (new version)) (aerobic, activated sludge, domestic, non-adapted) Readily biodegradable (according to OECD criteria).

Assessment of stability in water
According to structural properties, hydrolysis is not expected/probable.

Information on Stability in Water (Hydrolysis)
The product has not been tested. The statement has been derived from the structure of the product.

Bioaccumulative potential

Assessment bioaccumulation potential
Significant accumulation in organisms is not to be expected.

Bioaccumulation potential
Bioconcentration factor: 3.2, Fish (calculated)

Mobility in soil

Assessment transport between environmental compartments
The substance will not evaporate into the atmosphere from the water surface.
Adsorption to solid soil phase is not expected.

Additional information

Sum parameter

Chemical oxygen demand (COD): 350 mg/g

Biochemical oxygen demand (BOD) Incubation period 5 d: 175 mg/g

13. Disposal considerations

Waste disposal of substance:
Incinerate in suitable incineration plant, observing local authority regulations.

Container disposal:
Uncontaminated packaging can be recycled.

14. Transport Information

Land transport
TDG

Not classified as a dangerous good under transport regulations

Sea transport
IMDG

Not classified as a dangerous good under transport regulations

Air transport
IATA/ICAO

Not classified as a dangerous good under transport regulations

15. Regulatory Information

Federal Regulations

Registration status:

Chemical	DSL, CA	released / listed
Biocide	DSL, CA	blocked / not listed

NFPA Hazard codes:
Health: 2 Fire: 0 Reactivity: 0 Special:

Assessment of the hazard classes according to UN GHS criteria (most recent version):

Acute Tox.	4 (Inhalation - mist)	Acute toxicity
Skin Corr./Irrit.	2	Skin corrosion/irritation
Eye Dam./Irrit.	2B	Serious eye damage/eye irritation
Aquatic Acute	3	Hazardous to the aquatic environment - acute
Acute Tox.	5 (oral)	Acute toxicity
STOT SE	3 (irritating to respiratory system)	Specific target organ toxicity — single exposure
Skin Sens.	1	Skin sensitization

16. Other Information

SDS Prepared by:
BASF NA Product Regulations
SDS Prepared on: 2024/12/05

We support worldwide Responsible Care® initiatives. We value the health and safety of our employees, customers, suppliers and neighbors, and the protection of the environment. Our commitment to Responsible Care is integral to conducting our business and operating our facilities in a safe and environmentally responsible fashion, supporting our customers and suppliers in ensuring the safe and environmentally sound handling of our products, and minimizing the impact of our operations on society and the environment during production, storage, transport, use and disposal of our products.

END OF DATA SHEET